

SMARTBRIDGE / SKILL WALLET

Solution Requirements

Date: 2 July 2026

Team ID: —

Project Name: Caption AI : AI Powered Social Media Caption Generator

Define Functional and Non-Functional Requirements

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).

Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Med/Low)
1	Authentication	System loads environment secrets to authenticate backend routing sessions securely via local setup logic.	High
2	Authorization levels	Provides single-developer configuration dashboard control access for system administration.	Medium
3	External interfaces	Connects seamlessly to the Groq Cloud LLM API for lightning-fast inference text generation pipeline streaming.	High
4	Transactions processing	Processes incoming visual context inputs and text prompt fields into sanitized backend JSON payloads.	High
5	Reporting	Generates ready-to-copy social media text layouts populated with custom targeted hashtag categories.	High
6	Business rules, etc.	The system must automatically format outputs to match character constraints across distinct social profiles.	Medium
7	Compliance to laws	Adheres to isolated local storage principles to safeguard private user context data from unauthorized leakage.	Medium

S.No	Requirement Category	Requirement Description	Priority (High/Med/Low)
8	Other	Exposes localized text engines through secure proxy channels for external interface testing.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Text generation pipelines must process custom environment payloads swiftly without latency bottlenecks.	Inference response returns in < 2.5 seconds
2	Scalability	The architecture handles concurrent multi-platform character limit checks smoothly without system crashes.	Supports up to 50 concurrent local requests
3	Security & Data Privacy	Safeguards development parameters using runtime dot-env token injection setups.	Zero hardcoded backend API keys
4	Reliability & Availability	Local application micro-server routes must remain responsive throughout development sprint checks.	99.5% operational runtime availability
5	Usability & Accessibility	Provides a clear, readable text canvas interface that allows seamless copy-paste operations.	100% successful dashboard element rendering
6	Other	Ensures clean system components separation for trouble-free future feature expansions.	Decoupled frontend-backend logic layout